1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

Started on	Wednesday, 7 October 2026, 8:57 PM
State	Finished
Completed on	Thursday, 8 October 2026, 9:53 AM
Time taken	12 hours 56 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n-1;i++){
        for(int j=i+1;j<n;j++){
            if(arr[i]==arr[j]){
                printf("%d",arr[i]);
                break;
            }
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

Started on	Thursday, 8 October 2026, 9:54 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:02 AM
Time taken	7 mins 48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int r = 0;
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
        r ^= arr[i];
    }
    for(int i=1;i<n;i++){
        r ^= i;
    }
    printf("%d",r);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

Started on	Thursday, 8 October 2026, 10:32 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:41 AM
Time taken	8 mins 2 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int t;
    scanf("%d",&t);
    while(t--){
        int n;
        scanf("%d",&n);
        int arr[n];
        for(int i=0;i<n;i++){
            scanf("%d",&arr[i]);
        }
        int m;
        scanf("%d",&m);
        int brr[m];
        for(int i=0;i<m;i++){
            scanf("%d",&brr[i]);
        }

        for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            if(arr[i]==brr[j]){
                printf("%d ",arr[i]);
            }
        }
    }
}
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

Started on	Thursday, 8 October 2026, 10:41 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:52 AM
Time taken	10 mins 42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int main(){
    int t;
    scanf("%d",&t);
    while(t--){
        int n,m;
        scanf("%d",&n);
        int arr[n];
        for(int i=0;i<n;i++){
            scanf("%d",&arr[i]);
        }
        scanf("%d",&m);
        int brr[m];
        for(int i=0;i<m;i++){
            scanf("%d",&brr[i]);
        }
        int i=0,j=0;
        while(i<n && j<m){
            if(arr[i]==brr[j]){
                printf("%d ",arr[i]);
                i++;
                j++;
            }
            else if(arr[i]<brr[j]) i++;
            else j++;
        }
        printf("\n");
    }
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

5-Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

Started on	Thursday, 8 October 2026, 11:03 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:17 AM
Time taken	14 mins 28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *A = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    int k;
    scanf("%d", &k);

    int found = 0;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i != j && A[j] - A[i] == k) {
                found = 1;
                break;
            }
        }
        if (found) break;
    }

    printf("%d\n", found);

    free(A);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course

6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

Started on	Thursday, 8 October 2026, 11:18 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:27 AM
Time taken	8 mins 40 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    int *A = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &A[i]);
    }

    int k;
    scanf("%d", &k);

    int i = 0, j = 1;
    int found = 0;

    while (i < n && j < n) {
        if (i != j && A[j] - A[i] == k) {
            found = 1;
            break;
        } else if (A[j] - A[i] < k) {
            j++;
        } else {
            i++;
        }

        if (i == j) {
            j++;
        }
    }

    printf("%d\n", found);

    free(A);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course